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Asymmetric Information, Disagreement, and the Valuation of Debt and Equity

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1 Big picture

- **Question.** How do asymmetric information, investor disagreement, and liquidity trading jointly affect the valuation of debt and equity?
- **Main puzzle.** Empirical studies find that belief dispersion is negatively related to expected stock returns but positively related to expected bond returns. Traditional noisy rational expectations and disagreement models usually study securities with linear payoffs, so they cannot jointly explain these opposite relations for equity and debt.
- **Key mechanism.** Debt and equity are nonlinear claims on the same firm cash flow. Equity is convex in cash flows, while debt is concave. This payoff nonlinearity changes how prices aggregate private information and absorb liquidity trading shocks.
- **Information environment.** Investors receive dispersed private signals about firm cash flows and may disagree about how informative other investors' signals are. The disagreement parameter ρ governs whether investors fully learn from prices or partially dismiss the information in prices.
- **Liquidity trading.** Liquidity or noise traders submit demand shocks. These shocks reduce price informativeness and create risk compensation. The interaction of liquidity trading with nonlinear payoffs produces different expected-return predictions for debt and equity.
- **Baseline result.** With identical liquidity trading in debt and equity markets, the total value of debt plus equity equals the value of the unlevered firm. Modigliani–Miller holds in the baseline despite private information.
- **Expected-return result.** Expected equity and debt returns have opposite signs because equity prices are convex and debt prices are concave in the risk-neutral expectation of cash flows. The sign depends on whether the risk-neutral expectation is more or less volatile than a typical investor's posterior expectation.
- **Disagreement versus noisy prices.** Both disagreement and noisy price signals can generate belief dispersion, but they have opposite effects on expected returns. Noise-trading volatility tends to lower expected equity returns and raise expected debt returns; strong disagreement can reverse this pattern.
- **Capital-structure result.** When liquidity trading differs across debt and equity markets, debt and equity prices no longer perfectly aggregate the same information. Capital structure affects firm value even without taxes, distress costs, agency costs, or other classical frictions.
- **Systematic-risk result.** In a multi-firm model with a systematic risk factor, the private-information mechanism generates alphas relative to standard risk-factor benchmarks. It can explain low equity returns for distressed firms and positive bond return premia without relying only on systematic risk.
- **Takeaway.** The paper shows that asset-payoff nonlinearity is essential for understanding how belief dispersion, disagreement, and liquidity trading affect debt and equity returns.

2 Model primitives and measurement

2.1 Firm cash flow and security payoffs

The firm has total cash flow per share

$$\mathcal{V} = m + \theta, \quad \theta \sim N(0, \sigma_\theta^2). \quad (1)$$

The firm issues debt with face value K and equity. The payoffs are

$$V_E = \max\{\mathcal{V} - K, 0\}, \quad V_D = \min\{\mathcal{V}, K\}, \quad (2)$$

so that

$$V_E + V_D = \mathcal{V}. \quad (3)$$

Interpretation: Equity is a call option on firm cash flow, while debt is a concave claim that is capped at K . This nonlinearity is the central force in the paper.

2.2 Debt and equity supply

- The per capita supply of debt and equity is $\kappa \geq 0$.
- Setting the supply of each claim equal ensures that changes in capital structure do not mechanically change the total cash flow paid to investors.
- In the baseline model, liquidity traders submit identical orders $z \sim N(0, \sigma_z^2)$ in both debt and equity markets.

Interpretation: The identical liquidity-shock benchmark isolates payoff nonlinearity and information aggregation. Section 6 then relaxes the identical-liquidity assumption to study capital structure.

2.3 Investors, preferences, and terminal wealth

There is a unit mass of investors indexed by $i \in [0, 1]$. Each investor has CARA utility with risk tolerance τ over terminal wealth. Let $x_{E,i}$ and $x_{D,i}$ be demands for equity and debt. Terminal wealth is

$$W_i = \kappa(P_D + P_E) + x_{E,i}(V_E - P_E) + x_{D,i}(V_D - P_D). \quad (4)$$

Interpretation: CARA-normal structure makes demands tractable, but the payoff distribution is non-normal because debt and equity are nonlinear claims.

2.4 Private information

Each investor observes a private signal about the cash-flow shock:

$$s_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_\varepsilon^2). \quad (5)$$

Signals are conditionally independent across investors given θ .

Interpretation: Private information is dispersed across investors. Prices aggregate the cross-sectional information contained in order flow, but imperfectly because of liquidity trading and disagreement.

2.5 Disagreement parameter

The parameter $\rho \in [0, 1]$ captures the extent to which investors view other investors' signals as informative.

- $\rho = 1$: rational expectations. Investors fully recognize that other investors have useful information and therefore learn from prices.
- $\rho = 0$: pure disagreement or difference of opinions. Investors believe other investors' signals are not payoff relevant and do not learn from prices.
- $\rho \in (0, 1)$: partial disagreement. Investors learn from prices, but less than a rational-expectations investor would.

Interpretation: Belief dispersion can arise from two sources: noisy prices due to liquidity trading and disagreement about the informativeness of others' signals. The paper stresses that these two sources can have different pricing implications.

3 Equilibrium concepts and benchmark objects

3.1 Generalized linear equilibrium

Because debt and equity payoffs are nonlinear in cash flows, prices are not linear in fundamentals. The paper therefore uses a generalized linear equilibrium in which prices are invertible functions of a linear statistic of aggregate information and liquidity trading.

$$(P_D, P_E) = g(s_p), \quad (6)$$

where s_p summarizes aggregate private information and liquidity demand.

Interpretation: This equilibrium class is not a linear approximation. It allows prices to be nonlinear functions of a linear sufficient statistic, which is necessary because option-like payoffs generate nonlinear price functions.

3.2 Unlevered firm benchmark

For an unlevered firm, investor demand takes the standard form

$$x_i = \tau \frac{\mu_i - P_U}{\sigma_s^2}, \quad (7)$$

where $\mu_i = \mathbb{E}_i[\mathcal{V}]$ and σ_s^2 is posterior uncertainty. Market clearing implies

$$P_U = \int \mu_i di + \frac{\sigma_s^2}{\tau} (z - \kappa). \quad (8)$$

Interpretation: The unlevered price is a risk-neutral expectation of cash flow. It aggregates average investor beliefs and subtracts a risk premium for absorbing net supply and liquidity-trader demand.

3.3 Normal payoff functions

For $x \sim N(\mu_x, \sigma_x^2)$, define

$$M_E(\mu_x, \sigma_x^2, K) = \mathbb{E}[\max\{x - K, 0\}], \quad (9)$$

$$M_D(\mu_x, \sigma_x^2, K) = \mathbb{E}[\min\{x, K\}]. \quad (10)$$

Equity value is convex in μ_x , while debt value is concave in μ_x .

Interpretation: These functions are the normal-valuation analogs of call-option and capped-debt payoffs. Convexity and concavity drive the expected-return signs.

3.4 Risk-neutral uninformed benchmark

If investors are risk neutral and uninformed, prices are simply the expected payoffs:

$$P_E = M_E(m, \sigma_\theta^2, K), \quad P_D = M_D(m, \sigma_\theta^2, K). \quad (11)$$

Interpretation: The main model generalizes this benchmark by replacing the unconditional cash-flow mean with a risk-neutral expectation that depends on aggregate information and liquidity trade.

4 Baseline equilibrium with identical liquidity trading

4.1 Investor demands

Holding prices fixed, the paper shows that each investor's demands for debt and equity can be written as

$$\begin{pmatrix} x_{E,i} \\ x_{D,i} \end{pmatrix} = \frac{\tau}{\sigma_s^2} \left[\begin{pmatrix} \mu_i \\ \mu_i \end{pmatrix} - G \left(\begin{pmatrix} P_E \\ P_D \end{pmatrix} \right) \right], \quad (12)$$

for a function $G(\cdot)$ defined in the paper.

A key implication is

$$\frac{\partial x_{E,i}}{\partial \mu_i} = \frac{\partial x_{D,i}}{\partial \mu_i} = \frac{\tau}{\sigma_s^2}. \quad (13)$$

Interpretation: Even though equity is more sensitive to cash flows in some regions and debt is more sensitive in others, expected payoff and risk effects exactly offset. Investors speculate on their private beliefs with equal aggressiveness in debt and equity.

4.2 Proposition 1: equilibrium prices

In the baseline model, the equilibrium debt and equity prices are

$$P_E = M_E(P_U, \sigma_s^2, K), \quad P_D = M_D(P_U, \sigma_s^2, K), \quad (14)$$

where

$$P_U = \int \mu_i di + \frac{\sigma_s^2}{\tau} (z - \kappa). \quad (15)$$

Moreover,

$$P_E + P_D = P_U, \quad (16)$$

and equilibrium demands satisfy

$$x_{E,i} = x_{D,i} = \tau \frac{\mu_i - \int \mu_j dj}{\sigma_s^2} - z + \kappa. \quad (17)$$

Interpretation: The debt and equity prices equal expected payoffs under a risk-neutral normal distribution with mean P_U and variance σ_s^2 . Because both markets face identical liquidity trading, total firm value equals unlevered firm value; Modigliani–Miller holds in the baseline.

4.3 Corollary 1: comparative statics of prices

Debt and equity prices:

- increase in mean cash flow m ;
- increase in liquidity-trader demand z ;
- decrease in per capita supply κ .

The equity price decreases in the face value of debt K , while the debt price increases in K . Equity is convex in P_U ; debt is concave in P_U .

Interpretation: Favorable news and buy pressure have larger effects on equity than unfavorable news and sell pressure because equity is truncated from below. Debt displays the opposite curvature.

5 Expected returns on debt and equity

5.1 Return definitions

The paper works with dollar returns

$$R_E = V_E - P_E, \quad R_D = V_D - P_D. \quad (18)$$

When payoff functions are linear, expected returns are driven mainly by net supply. With nonlinear claims, expected returns can be nonzero even when per capita supply is zero.

Interpretation: Nonlinear payoffs make idiosyncratic information matter for expected returns even when the firm is in zero net supply.

5.2 Proposition 2: variance comparison

Assume zero per-capita supply, $\kappa = 0$. Expected equity returns are positive if and only if

$$\text{Var}[P_U] < \text{Var}_i[\mu_i], \quad (19)$$

while expected debt returns are positive if and only if

$$\text{Var}[P_U] > \text{Var}_i[\mu_i]. \quad (20)$$

The reason is that

$$\mathbb{E}[R_E] = \mathbb{E}_i[M_E(\mu_i, \sigma_s^2, K)] - \mathbb{E}[M_E(P_U, \sigma_s^2, K)], \quad (21)$$

$$\mathbb{E}[R_D] = \mathbb{E}_i[M_D(\mu_i, \sigma_s^2, K)] - \mathbb{E}[M_D(P_U, \sigma_s^2, K)]. \quad (22)$$

Interpretation: Since M_E is convex, Jensen's inequality makes expected equity returns depend positively on the volatility of the typical investor's posterior expectation relative to the risk-neutral expectation. Since M_D is concave, debt has the opposite sign.

5.3 Role of liquidity-trading volatility

Liquidity-trading volatility affects $\text{Var}[P_U]$ because prices must compensate investors for absorbing liquidity trader demand. When σ_z^2 is high, P_U becomes more volatile than individual investor beliefs.

- Higher σ_z^2 tends to reduce expected equity returns.
- Higher σ_z^2 tends to increase expected debt returns.
- This jointly matches evidence that belief dispersion is negatively related to stock returns but positively related to bond returns when dispersion is driven by noisy prices.

Interpretation: Price noise raises belief dispersion but also increases the volatility of risk-neutral cash-flow expectations, making debt returns higher and equity returns lower.

5.4 Role of disagreement

The disagreement parameter ρ changes how much investors learn from prices.

- With rational expectations ($\rho = 1$), investors fully learn from the price signal.
- With strong disagreement (ρ low), investors underweight or ignore information in prices.
- Strong disagreement can make $\text{Var}_i[\mu_i]$ large relative to $\text{Var}[P_U]$.

Interpretation: Disagreement raises belief dispersion but can increase expected equity returns and decrease expected debt returns. This is the opposite of the effect of noise-trading volatility. Therefore belief dispersion is not a sufficient statistic; its source matters.

5.5 Default risk

The model predicts that expected returns can vary with default probability in nonstandard ways.

- Expected equity returns can be hump-shaped in default risk.
- Expected debt returns typically increase with default risk for investment-grade debt.
- For firms close to bankruptcy, the relation can reverse because equity and debt payoff curvature changes near the default boundary.

Interpretation: This can rationalize empirical evidence on the distress anomaly in equity markets and credit-risk premia in bond markets without relying solely on systematic risk.

6 Differing liquidity trade in debt and equity markets

6.1 Motivation

The baseline assumes identical liquidity trading in debt and equity. Section 6 relaxes this by allowing liquidity demand to be a bivariate normal vector

$$z = (z_E, z_D), \quad z \sim N(0, \Sigma_z). \quad (23)$$

Interpretation: In reality, debt and equity markets differ in clientele, liquidity, market structure, and noise-trader intensity. Allowing different liquidity shocks makes capital structure matter.

6.2 Generalized equilibrium

With different liquidity trading across markets, prices take the form

$$(P_D, P_E) = g(s_{p1}, s_{p2}), \quad (24)$$

where the two price signals separately summarize information from debt and equity order flow.

Interpretation: Once liquidity shocks differ across debt and equity, each market can reveal different information. Debt and equity prices no longer collapse to a single sufficient statistic.

6.3 Cross-market demand spillovers

When liquidity traders buy equity, equity prices respond directly, but debt prices can also move because investors can substitute across the two securities.

- If liquidity shocks are highly correlated, demand spillovers are stronger.
- If markets are segmented, price reactions become market-specific.
- The more nonlinear the payoff region, the more leverage changes the spillover pattern.

Interpretation: Cross-market spillovers depend on how debt and equity jointly span exposure to the firm's total cash flow.

6.4 Debt-equity price correlation

The model predicts that the correlation between debt and equity prices is maximized when default probability is around 50%.

- Far from default, debt is close to risk-free and moves little with cash-flow news.
- Close to bankruptcy, equity is close to worthless and moves little with cash-flow news.
- Around the default boundary, both claims are sensitive to cash-flow news.

Interpretation: Payoff curvature creates a non-monotonic relation between leverage and debt-equity price correlation.

7 Capital structure and firm valuation

7.1 Modigliani–Miller in the baseline

With identical liquidity trading across the two markets,

$$P_E + P_D = P_U, \quad (25)$$

so capital structure is irrelevant for total firm value.

Interpretation: The baseline deliberately preserves the Modigliani–Miller result even under private information. This makes the later violations transparent.

7.2 Violations with different liquidity trading

When liquidity trading differs across debt and equity, total firm value becomes

$$\mathbb{E}[P_E + P_D] \neq \mathbb{E}[P_U]. \quad (26)$$

If liquidity trading is more intense in the equity market than in the debt market, leverage can raise total firm value by shifting some claims into the less liquidity-noisy debt market.

Interpretation: Capital structure affects valuation through market microstructure and information aggregation rather than taxes, distress costs, or agency costs.

7.3 Optimal capital structure

The firm can choose the face value K of debt to maximize

$$\mathbb{E}[P_E(K) + P_D(K)]. \quad (27)$$

The optimal leverage depends on:

- the relative liquidity-trading volatility in debt and equity markets;
- the correlation of liquidity shocks across markets;
- the degree of disagreement ρ ;
- the location of the default boundary relative to the cash-flow distribution.

Interpretation: Even without classical capital-structure frictions, there can be an interior optimal capital structure because securities have different information and liquidity characteristics.

8 Systematic risk and multiple firms

8.1 Cash-flow structure

The paper extends the model to many firms and a common factor. Firm n has cash flow

$$\mathcal{V}_n = m_n + \beta_n F + \theta_n, \quad (28)$$

where F is a systematic factor and θ_n is idiosyncratic.

Interpretation: This extension separates systematic risk, which is priced in standard asset-pricing models, from the private-information and liquidity-trading forces studied in the baseline model.

8.2 Factor asset

A factor asset is introduced so that investors can hedge systematic risk. This allows the paper to study alphas relative to the factor benchmark.

Interpretation: If private-information effects survive after controlling for systematic risk, they can explain expected-return patterns that look like asset-pricing anomalies.

8.3 Expected returns with systematic risk

The model generates expected returns that combine:

- standard risk premia from exposure to the common factor;
- idiosyncratic information premia caused by nonlinear debt and equity payoffs;
- liquidity-trading premia from absorbing order-flow shocks.

Interpretation: Expected debt and equity returns can display patterns usually interpreted as anomalies even after controlling for systematic risk exposures.

8.4 Relation to distress risk and credit spreads

The model predicts:

- expected equity returns may decrease with distress risk for many firms;
- expected debt returns may increase with default risk for investment-grade debt;
- both relations can reverse near bankruptcy;
- debt-equity return correlations are low when default risk is very low or very high and higher around the middle of the default distribution.

Interpretation: This aligns with evidence on the distress anomaly, credit spreads, and equity-bond comovement.

9 Empirical implications

9.1 Belief dispersion and returns

The paper's first empirical message is that belief dispersion must be decomposed by source.

- Dispersion from noisy prices or liquidity trading should be associated with lower equity returns and higher debt returns.

- Dispersion from disagreement can generate the opposite sign pattern.
- Empirical measures such as analyst forecast dispersion, CDS disagreement, bond analyst dispersion, and option-implied disagreement may capture different mixtures of the two sources.

Interpretation: The same empirical variable called “belief dispersion” can map into different theoretical objects.

9.2 Default risk and expected returns

The model predicts nonlinear relations:

- expected equity returns are hump-shaped in default risk;
- expected debt returns usually increase with default risk but can reverse near bankruptcy;
- the sign and magnitude depend on disagreement and liquidity-trading intensity.

Interpretation: This helps reconcile evidence that distressed equities have low returns while risky bonds have high returns.

9.3 Capital structure and valuation

When debt and equity markets differ in liquidity trading, leverage affects firm value.

- Leverage can raise firm value if debt market liquidity shocks are smaller than equity market shocks.
- Leverage can reduce firm value in the opposite case.
- Optimal capital structure depends on information quality and market liquidity, not only tax shields and distress costs.

Interpretation: The model provides a microstructure-based valuation channel for capital structure.

9.4 Bond-equity correlations

The model predicts that bond-equity return correlations vary with default risk.

- Correlations are low for very safe firms because debt behaves like a risk-free bond.
- Correlations are low for near-bankruptcy firms because equity is close to an out-of-the-money option.
- Correlations are higher in the middle because both claims are sensitive to cash-flow news.

Interpretation: This offers an explanation for empirical non-monotonicity in stock-bond correlations across credit-quality groups.

10 Figures and original visual evidence

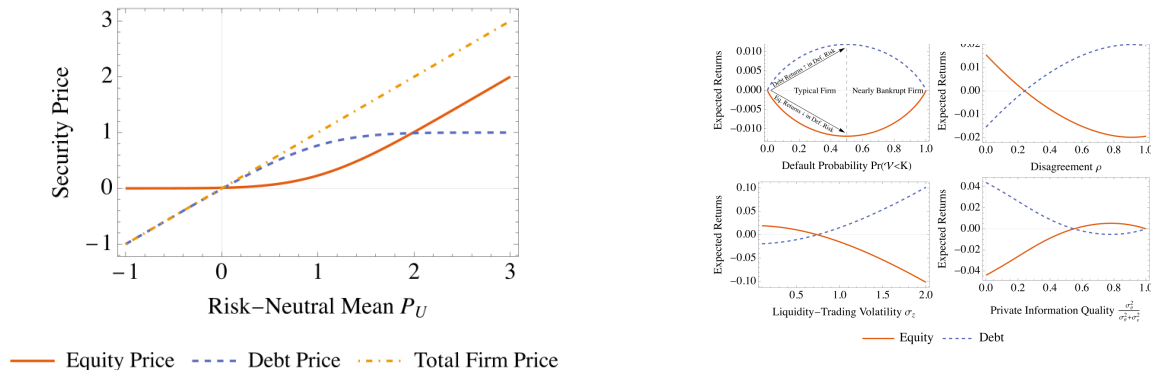
The paper is theoretical and contains no empirical tables. The visuals below are directly cropped from the paper and grouped compactly.

10.1 Figure 1: debt and equity price functions

Read:

- The equity price is convex in the risk-neutral cash-flow expectation P_U .
- The debt price is concave in P_U .
- The sum of debt and equity prices equals the unlevered price in the baseline.

Economic message: This is the key geometry of the model. Convexity for equity and concavity for debt convert changes in the volatility of beliefs into opposite expected-return effects.



10.2 Figure 2: expected return comparative statics

Read:

- The figure shows expected equity and debt returns as functions of leverage, private information quality, disagreement, and liquidity-trading volatility.
- Equity and debt returns move in opposite directions across many parameter changes.
- Default-risk relations are nonlinear, with reversals near extreme leverage.

Economic message: The figure illustrates the central prediction: the same source of belief dispersion can affect equity and debt returns in opposite ways because of payoff curvature.

10.3 Figures 3 and 4: cross-market spillovers and debt-equity correlation

Read:

- Figure 3 shows how equity and debt prices respond to liquidity trading in each market.
- Liquidity shocks in one market spill over to the other through investors' portfolio choices.
- Figure 4 shows that debt-equity price correlation is non-monotonic in default risk.

Economic message: Debt and equity markets are linked by a common cash-flow fundamental but separated by payoff nonlinearity and liquidity trading. This generates asymmetric cross-market reactions.

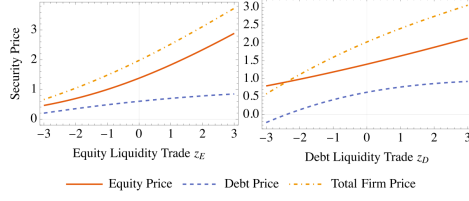


Figure 3: cross-market demand spillovers

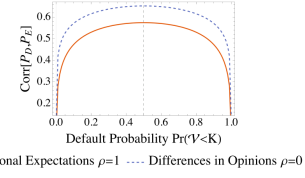


Figure 4: leverage and debt-equity price correlation

10.4 Figures 5 and 6: violations of Modigliani–Miller and optimal capital structure

Read:

- Figure 5 shows that total expected firm value can differ from the value of the unlevered firm when debt and equity liquidity trading differ.
- Figure 6 shows that firm value can be maximized at an interior leverage level.
- The optimal leverage depends on which market has more liquidity-trading volatility and on the degree of disagreement.

Economic message: Capital structure becomes valuable because it changes how cash-flow exposure is distributed across markets with different information and liquidity properties.

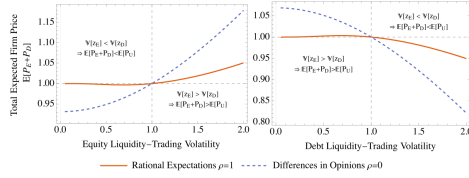


Figure 5: violations of Modigliani–Miller

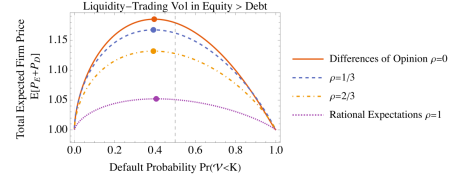


Figure 6: optimal capital structure

10.5 Figure 7: expected returns with systematic risk

Read:

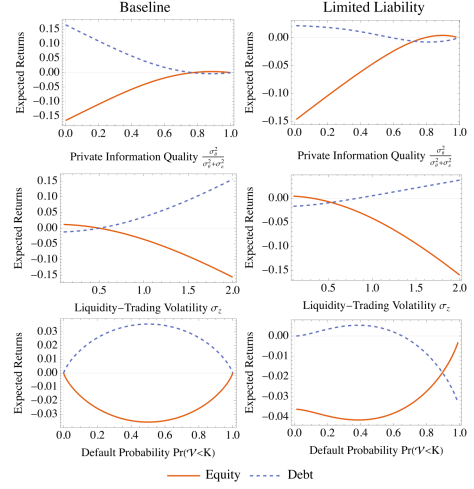
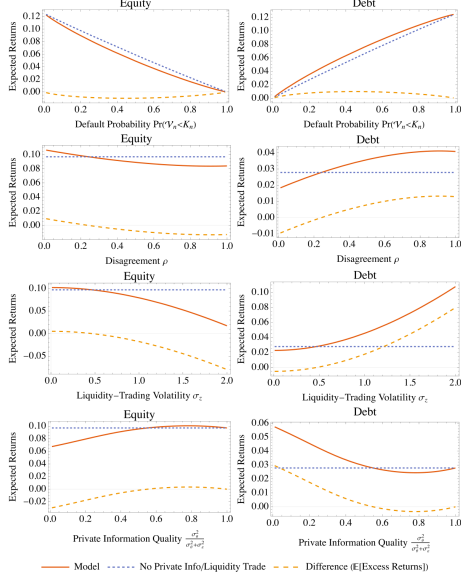
- Figure 7 adds systematic risk to the model and compares model-implied expected returns with a benchmark without private information or liquidity trade.
- The figure shows that the private-information mechanism can produce alphas relative to standard risk-factor pricing.
- Distress-risk relations differ for equity and debt and can reverse for firms close to default.

Economic message: The private-information channel survives in a multi-firm setting with systematic risk. The model therefore speaks to empirical asset-pricing anomalies rather than only to idiosyncratic toy examples.

10.6 Appendix Figure D.8: limited liability robustness

Read:

- The appendix modifies the model so total firm cash flow respects limited liability.
- The qualitative comparative statics remain similar to the baseline model.
- Private information quality, liquidity-trading volatility, and default risk continue to affect expected debt and equity returns in opposite ways.



Economic message: The main results are not an artifact of allowing debt payoffs to become negative in the baseline normal-cash-flow setup.

11 Short summary

- The paper studies debt and equity valuation when investors have private information and may disagree about the informativeness of others' signals.
- Debt and equity are nonlinear claims on the same cash flow; equity is convex and debt is concave.
- With identical liquidity trading in the two markets, total firm value equals the unlevered value, so Modigliani–Miller holds.
- Expected returns on debt and equity depend on the difference between the volatility of the risk-neutral expectation and the volatility of a typical investor's posterior expectation.
- Equity and debt expected returns typically move in opposite directions.
- Belief dispersion from noisy prices and belief dispersion from disagreement can have opposite implications.
- Default risk affects expected debt and equity returns nonlinearly; relations can reverse near bankruptcy.
- When liquidity trading differs across debt and equity markets, capital structure affects total firm value.
- Leverage can raise firm value if it shifts cash-flow exposure toward a market with lower liquidity-trading volatility.
- A multi-firm extension shows that these mechanisms survive after adding systematic risk.
- The model helps reconcile evidence on analyst dispersion, bond return premia, distress equity returns, capital structure, and debt-equity correlations.

12 Conclusion

12.1 Core conclusion

Banerjee, Breon-Drish, and Smith show that the valuation of debt and equity under heterogeneous beliefs cannot be understood from linear-payoff models alone. Because equity and debt are nonlinear claims, asymmetric information, disagreement, and liquidity trading have different effects across the two securities.

12.2 Economic intuition

The core intuition is Jensen's inequality applied to security payoffs. Equity is convex in the risk-neutral cash-flow expectation, so it benefits when a typical investor's posterior expectation is more volatile than the risk-neutral expectation. Debt is concave, so it benefits in the opposite case. Liquidity trading and disagreement determine which expectation is more volatile.

12.3 Takeaway

The paper provides a unified theory of how belief dispersion affects stocks and bonds. It explains why the same empirical object, belief dispersion, can be associated with low equity returns and high bond returns. It also shows that capital structure can affect firm value through information aggregation and liquidity trading, even in the absence of standard capital-structure frictions.